Online Retail II — Cancellation Loss Analysis

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Business Problem Definition

Return rates are an increasing operational cost for e-commerce. The average return rate has more than doubled since 2019, from 8.1 to 16.9 percent in 2024, and consumers worldwide have returned over $890 billion worth of merchandise (National Retail Federation & Happy Returns, 2024). Processing a single return can be between 20% and 65% of the item's original price, before you add labor to restock the product and account for lost customer lifetime value. Return rates also increase by about 17 percent above the yearly average during the holiday season when business is at its peak.

Most companies do not know how returns occur or why they happen. This study will analyze cancellation patterns by customer demographics, seasonal trends, geographic location, and product type to identify where most of these losses occur and the reasons for those losses.

Objectives per Section

Geographic and Temporal

- Are return rates significantly higher in specific countries?
- Do return volumes vary across months or quarters, indicating a seasonal pattern?

Customer Return Risk

- What share of customers accounts for the majority of return volume?
- Do high-return customers show distinct purchasing behavior, such as higher order frequency or larger basket sizes?
- Is there a correlation between purchase frequency and return rate?

Product Return Risk

- Which products have the highest return rates as a share of units sold?
- Do high-return products share characteristics such as price range or category?
- Does a Pareto pattern hold? Do a small number of products account for most returns?

KPIs Used in this Project

KPI	Definition	Unit	Used In
Cancellation Rate	Share of invoices canceled out of total invoices placed	%	Geographic & Temporal
Cancellation Loss	Total monetary value of canceled orders	£	All sections
Pareto Concentration	Number of customers or SKUs needed to reach 80% of total cancellation loss	Count	Customer & Product
Recency (R)	Days since a customer's last cancellation, measured from the dataset end date	Days	Customer
Frequency (F)	Total unique purchase invoices placed by a customer across the full dataset	Count	Customer
Monetary (M)	Total gross purchases in £ across the full dataset, excluding cancellations	£	Customer
Cancellation Loss per Customer (Re)	Total cancellation loss in £ attributed to a specific customer	£	Customer
Return Rate by SKU	Share of units canceled relative to total units sold for a given product	%	Product

Methods and Tools

Data

The analytical data is taken from the Online Retail II Dataset at the UCI Machine Learning repository. This dataset contains over one million (1,048,576) individual transaction records for an online UK-based retailer. There were eight columns of information in the dataset: Invoice #, Stock Code, Description, Quantity purchased, Invoice date, Price per item, Customer ID, and Country of purchase.

Two data signals can clearly identify cancellation events. These include all invoice numbers that begin with 'C', as well as those in which a positive quantity indicates the return of goods. As such, no additional assumptions have been made beyond what is found within the data itself.

Cleaning decisions:

- Zero and Negative Price Rows Removed any potential errors from the Data Entry process that do not have an Analytical Value.
- Administrative Non-Product Codes (Manual/D/Discount/POST) are excluded. The Administrative Codes do not provide Product or Customer Signal.
- SKU Code M (Manual Adjustments) Retained within Shared Dataset to allow Geographic & Customer Analysis. Regardless of whether the product was on the item, the dollar amount lost is real; however, we excluded SKU-level analysis from our Products Section because it would bias the product Information section.

Analytical Techniques

Geographic and Temporal

Technique	Justification
Country-level aggregation	The loss (in £) and cancellation rate (%) were calculated as separate metrics. It may be possible for a market to have a high cancellation rate but low overall financial loss or vice versa. Therefore, it would be misleading to combine these two metrics when allocating intervention.
Chi-square test of independence	This test assesses whether the difference in cancellation rates between the UK and other markets is due to chance or reflects a real statistical difference. It prevents the conclusion that observed rate differences are merely random occurrences of variation rather than real differences.
Time series analysis (monthly and quarterly)	A time-series analysis at a monthly level captures potential spike events in each month. In contrast, a quarterly analysis allows trends in cancellation rates across seasons to emerge, which may be obscured by the "noise" of cancellations across individual months.
Linear trend fitting	. To quantify whether the Cancellation Rate is trending up or down. A visual interpretation of the Trend Line would allow for subjective interpretations.
Year-over-year comparison	To separate losses attributable to increased Volume from those representing actual Behavioral Improvement. If Losses decrease but CR remains constant, this indicates that the Business is selling less.

Customer

Technique	Justification
Pareto analysis (80% threshold)	Finds the fewest accounts responsible for most of the losses in each country. Unlike the top-N method, this method does not impose an arbitrary cutoff; instead, it adjusts to each market's concentration structure.
Descriptive statistics (min, Q25, median, mean, Q75, max)	This methodology is used as a pre-Pareto distributional diagnosis. The fact that the mean is much larger than the median confirms the presence of heavy tails (right skew) and therefore supports using concentration-based segmentation methods over those based on averages.
RFMR framework	The standard RFM model has been adapted to account for cancellation risk. In F, purchase frequency is measured by total purchases rather than cancellation frequency to prevent frequent customers from being penalized. Raw cancellation loss in pounds (£), rather than a proportion, is used in Re to better assess the risk level of small, one-time buyers with a very high but financially small return rate.
Quintile scoring (1-5)	Each dimension in RFMR is scored on a scale of 1 to 5 to ensure that scores represent relative risk levels across markets rather than a single threshold.
Spearman rank correlation	Determines whether the relationship between an individual's number of purchases and the percentage of returns they make is negative. Since the distributions contain many large values, resulting in right-skewness, and include many outlying observations, Spearman rank correlation was selected over Pearson, as Spearman ranks all observations before calculating the correlation, thereby allowing it to address both issues.

Product

Technique	Justification
Pareto analysis at the SKU level	Identifies the customer segment to enable easy comparison of dimension concentration levels by mirroring it. When both are consistent with a Pareto distribution, the business is facing cumulative (or compounding) risk.
Price band segmentation	Aggregates SKUs into six price bands and independently calculates total loss and average return rates. The two measures provide different stories about product risks - you cannot get a complete view of where the product risks reside from reviewing only one measure.
Event-level invoice drill-down	Determines whether spikes represent system-wide problems or singular events by tracing data back to invoices, customers, and individual SKUs. Only aggregate-level analysis cannot be used to differentiate.
Key account product breakdown	Examines if multiple accounts are canceling the same products under a Pareto model -- this is an indicator of Product-Market Fit versus when a single account cancels many different SKUs in a single event -- this indicates poor Account Management as opposed to needing Product Intervention.

Tools

Tool	Purpose
Python: pandas, numpy	Data loading, cleaning, transformation, aggregation
Python: matplotlib	Visualizations
Python: scipy.stats	Chi-square tests and Spearman correlation
Jupyter Notebook	Analysis environment
GitHub	Version control and local execution

Analysis and Results

Geographic Analysis

Question 1: Are return rates significantly higher in specific countries?

Yes. Six countries (the UK, Eire, Germany, France, Spain & Norway) account for approximately 95% of the total cancellation loss. The six-country group represents a substantial portion of the financial exposure and sufficient data volume to draw conclusions that can be reasonably relied upon.

Approximately 85.2% of total cancellation losses were in the United Kingdom. Germany & Ireland both had significantly higher cancellation rates than the United Kingdom, but lower order volumes. A chi-square test showed that rate differences between the U.K. & all other countries are statistically different except for Sweden.

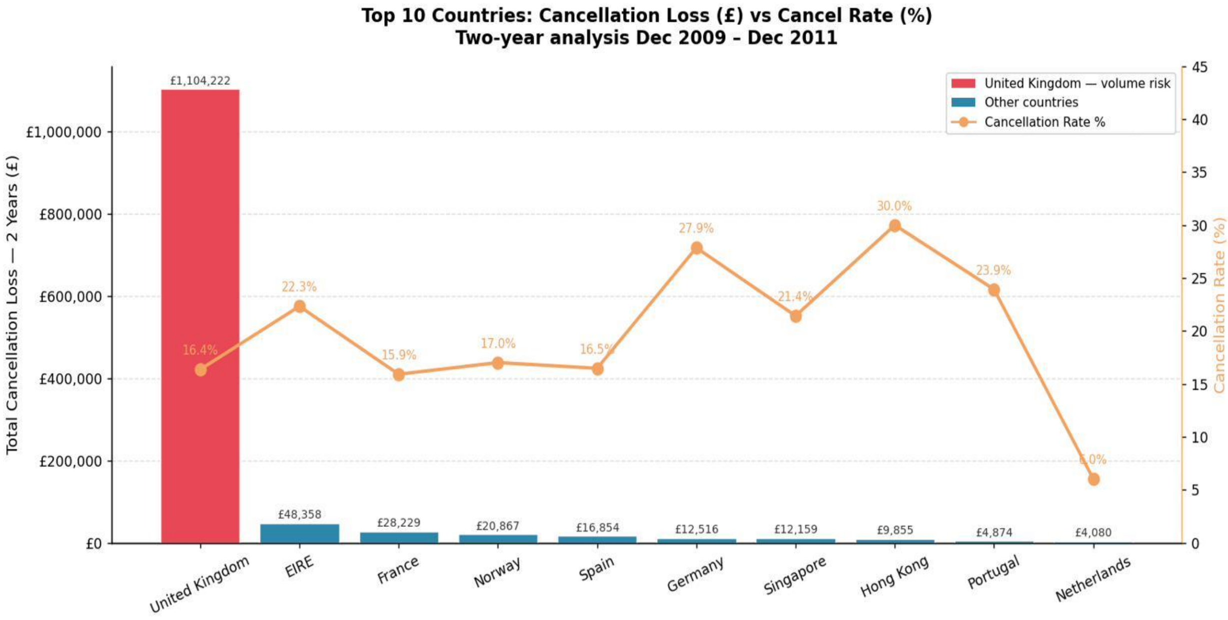

Figure 1: Top 10 Countries — Cancellation Loss (£) vs Cancellation Rate (%)

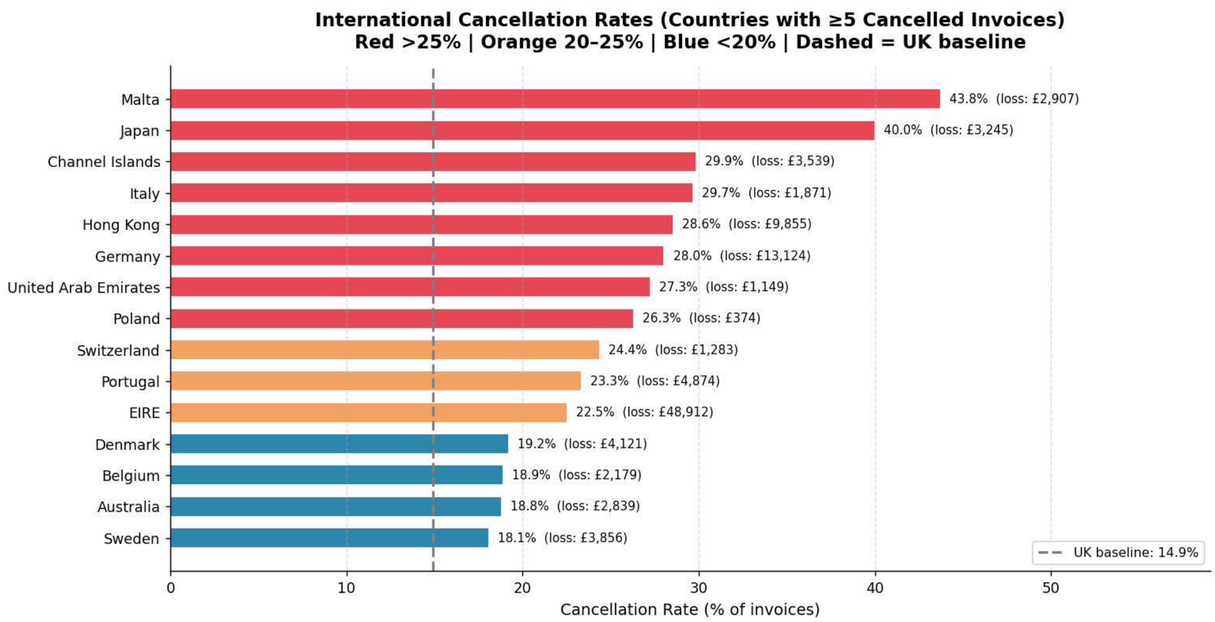

Figure 2: International Cancellation Rates — Countries with 5 or More Canceled Invoices

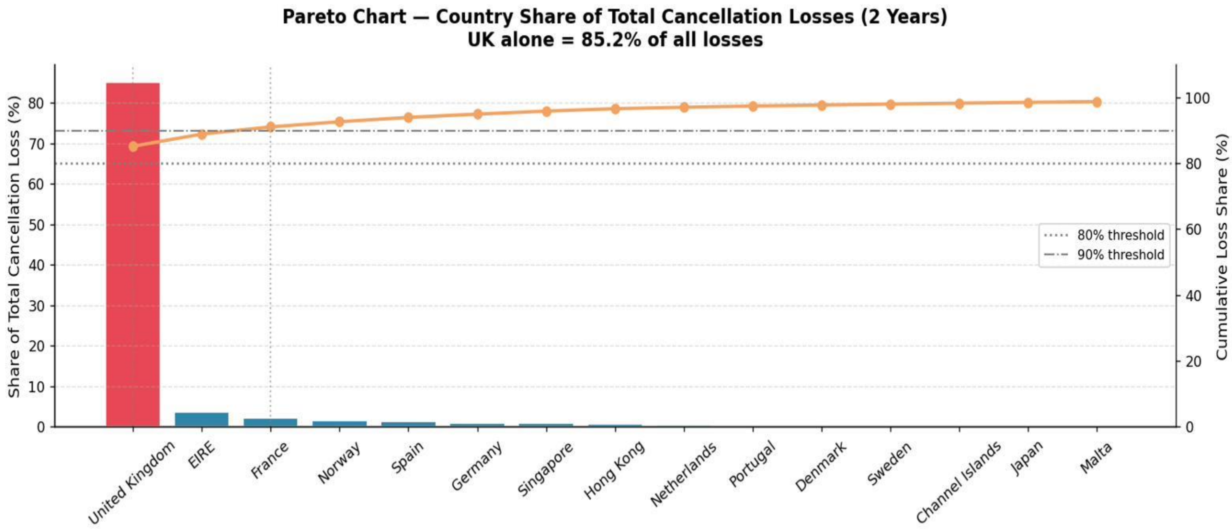

Figure 3: Pareto Chart — Country Share of Total Cancellation Losses

Business Insight: The UK is the single largest exposure. But fixing the UK does not fix the rate problem — Germany and EIRE have higher rates that require a different approach.

Temporal Analysis

Question 2: Do return volumes vary across months or quarters, pointing to seasonal patterns?

Yes. The monthly analysis of the data identified three major spikes in cancellations (losses) in September 2010, December 2010, and January 2011. Quarterly analysis revealed that losses will be greater in the fourth quarter and at the beginning of the first quarter than at other times of the year.

The cancellation rate was relatively constant throughout this period, averaging about 16.9 percent over two years. The spikes represent an increase in total cancellations due to higher fourth-quarter volumes, rather than a change in cancellation behavior.

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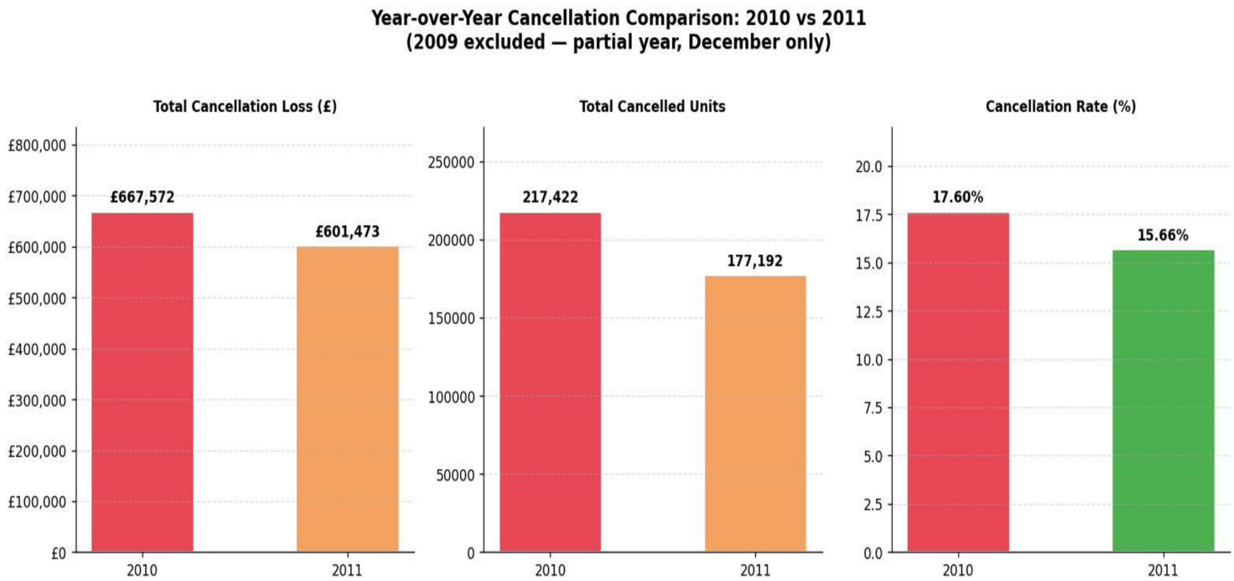

Figure 4: Year-over-Year Cancellation Comparison — 2010 vs 2011

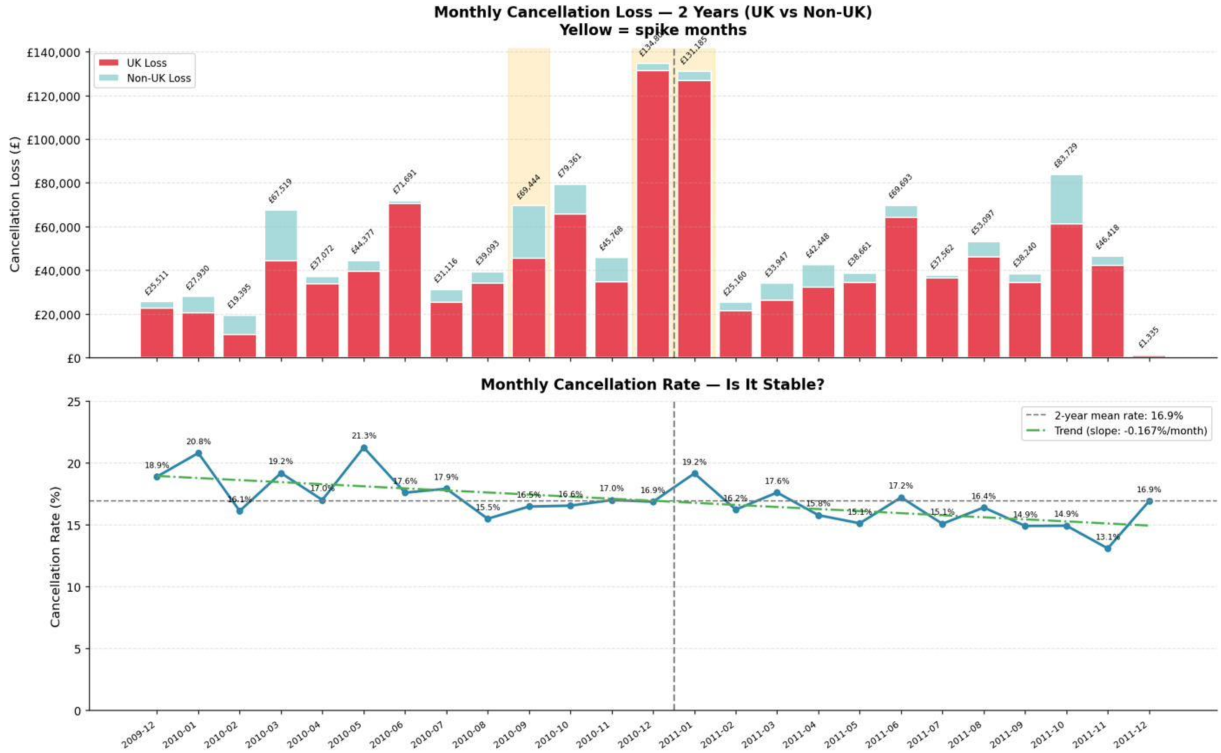

Figure 5: Monthly Cancellation Loss — 2 Years (UK vs Non-UK)

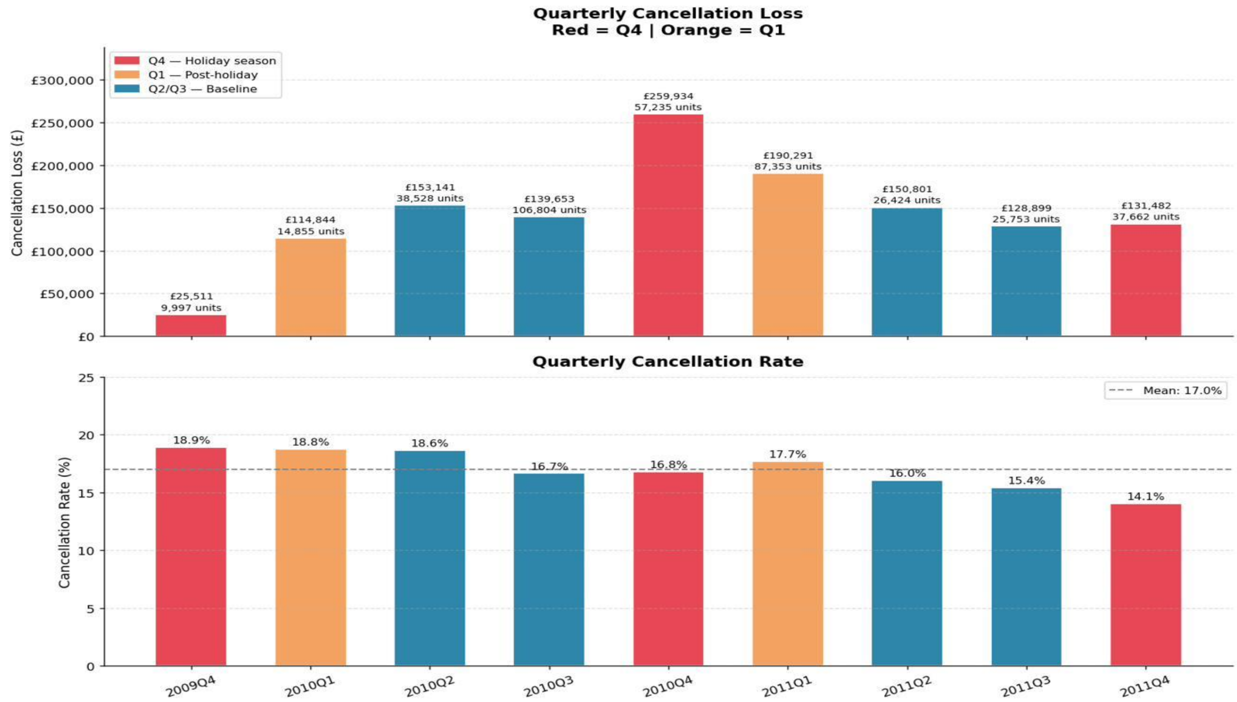

Figure 6: Quarterly Cancellation Loss and Rate

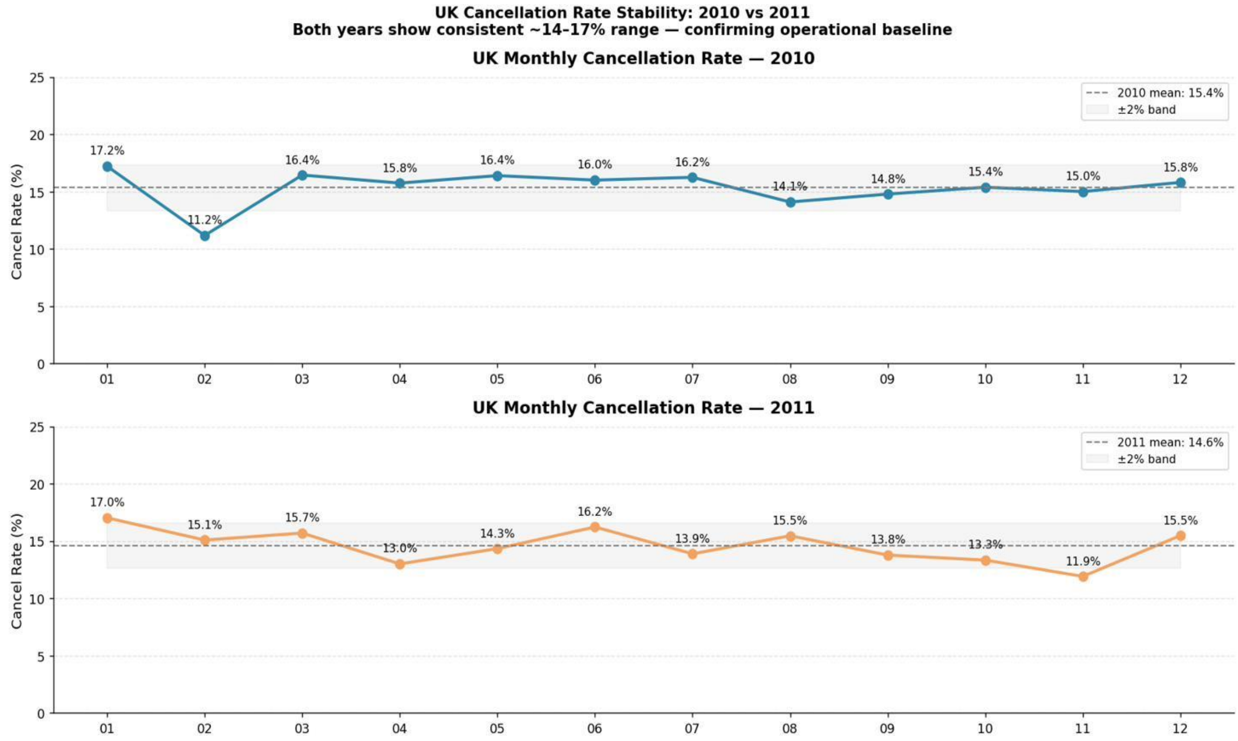

Figure 7: UK Cancellation Rate Stability — 2010 vs 2011

Business Insight: Pre-position logistics and customer service resources before Q4 and Q1.

Customer Analysis

Question 1: What share of customers accounts for the majority of return volume?

Customer-level losses were aggregated across the top six markets. Before identifying which customers drive those losses, a distribution analysis of cancellations per invoice was conducted. The results were a histogram that revealed extreme right-skewness in the loss distribution per invoice. That is, median losses were £16.95, mean losses were £160.65, and maximum losses reached £77,184. The gap between the two demonstrates why average-based analyses would misrepresent the typical cancellation event and justifies the use of Pareto analysis.

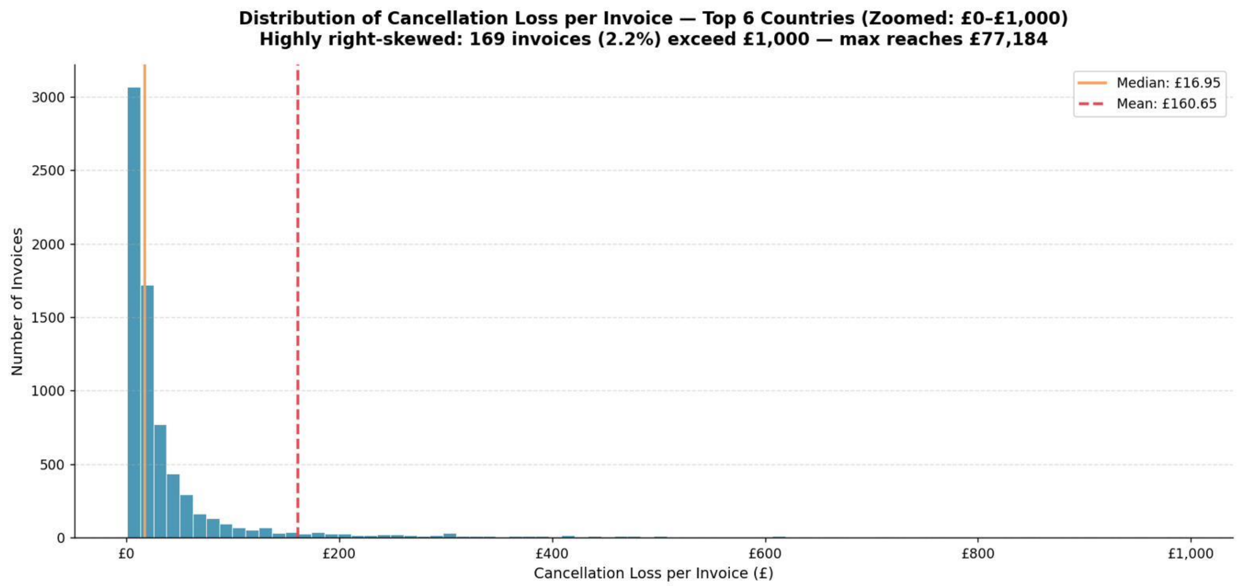

Figure 9: Distribution of Cancellation Loss per Invoice — Top 6 Countries (Zoomed: £0-£1,000)

Pareto analysis indicates significant differences in concentration among the six markets examined.

- EIRE: Two accounts account for virtually all of the losses experienced. Specifically, customer 14911 at 59.12% (£25,169) and Customer 14156 at 40.83% (£17,381).
- Norway: one account alone experiences 94.47% of total losses; specifically, customer 15760 (£19,712). Most of these represent real monetary losses with no product detail attached. StockCode M manual entries.
- France: Three accounts experience 81.20%. Of these, Customer 14277 at 45.45% (£12,829), Customer 12536 at 30.09% (£8,495), and Customer 12589 at 5.66% (£1,598).
- Spain: only one account has more than half of all losses (£12,387); specifically, customer 12454. A second account (customer 12539) has another 10.18% of losses.
- Germany: nineteen accounts required to reach 80% exposure (of a base of 66). Thus, while there are dominant accounts within Germany, no single account is dominant. The largest account holds only 14.56%.
- UK: ninety-nine percent of all accounts (only ten accounts remain unaccounted for out of 2,292 accounts) must be clustered or otherwise grouped to address at scale. Customer 12346 leads at 10.77% (£77,620) and Customer 15098 at 5.45% (£39,267).

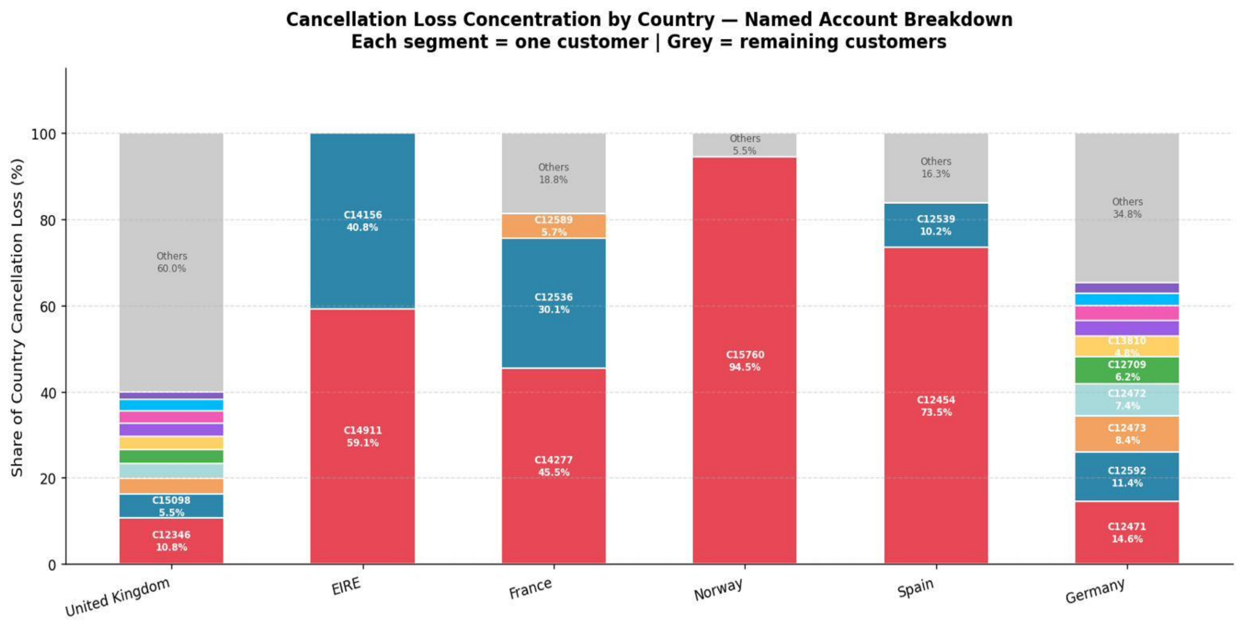

Figure 11: Cancellation Loss Concentration by Country — Named Account Breakdown

Business Insight: Four countries/markets (EIRE, Norway, France, and Spain) are characterized by having fewer than 10 accounts that explain the majority of exposure in each market. Germany and the UK differ from the other four countries. In both cases, the long tail of retail accounts suggests a pattern that requires segmentation before intervention.

Question 2: Do high-return customers show distinct purchasing behavior, such as higher order frequency or larger basket sizes?

The RFMR framework was developed for both the 190 UK and 19 Germany Pareto accounts. Each account has been rated against four criteria:

- R (Recency): Days since the customer's last cancellation and through the data set end date (Dec. 9, 2011). Recency is inverted in the rating process, so that a customer who most recently canceled receives a score of 5.

- F (Frequency): Total number of unique invoices purchased by the customer over the entire data set. This captures all of the customer's purchasing history rather than just their cancellation history.

- M (Monetary): Total pounds spent by the customer over the entire data set. Monies spent on cancellations have been excluded from this total. This measure represents the customer's financial influence as a buyer.

- Re (Cancellation Loss): Total monetary value lost due to cancellations. A ratio has not been used to calculate cancellation loss as it skews the risk assessment of a customer based upon the proportionate rate they incur as opposed to the actual financial loss incurred by the customer.

Each of the four dimensions has been ranked on a scale of 1–5 via quintile binning. Within each country, 1 represents the lowest level of risk while 5 indicates the highest level of risk. Quintile binning was chosen to rank accounts within each country because, although multiple bins are possible, there will always be fewer bins than accounts in an account base. It is also worth noting that accounts in Ireland, France, Norway, and Spain were excluded from analysis. With only 1–3 major accounts per country, there are too few accounts to achieve statistical significance when ranking accounts using quintile binning.

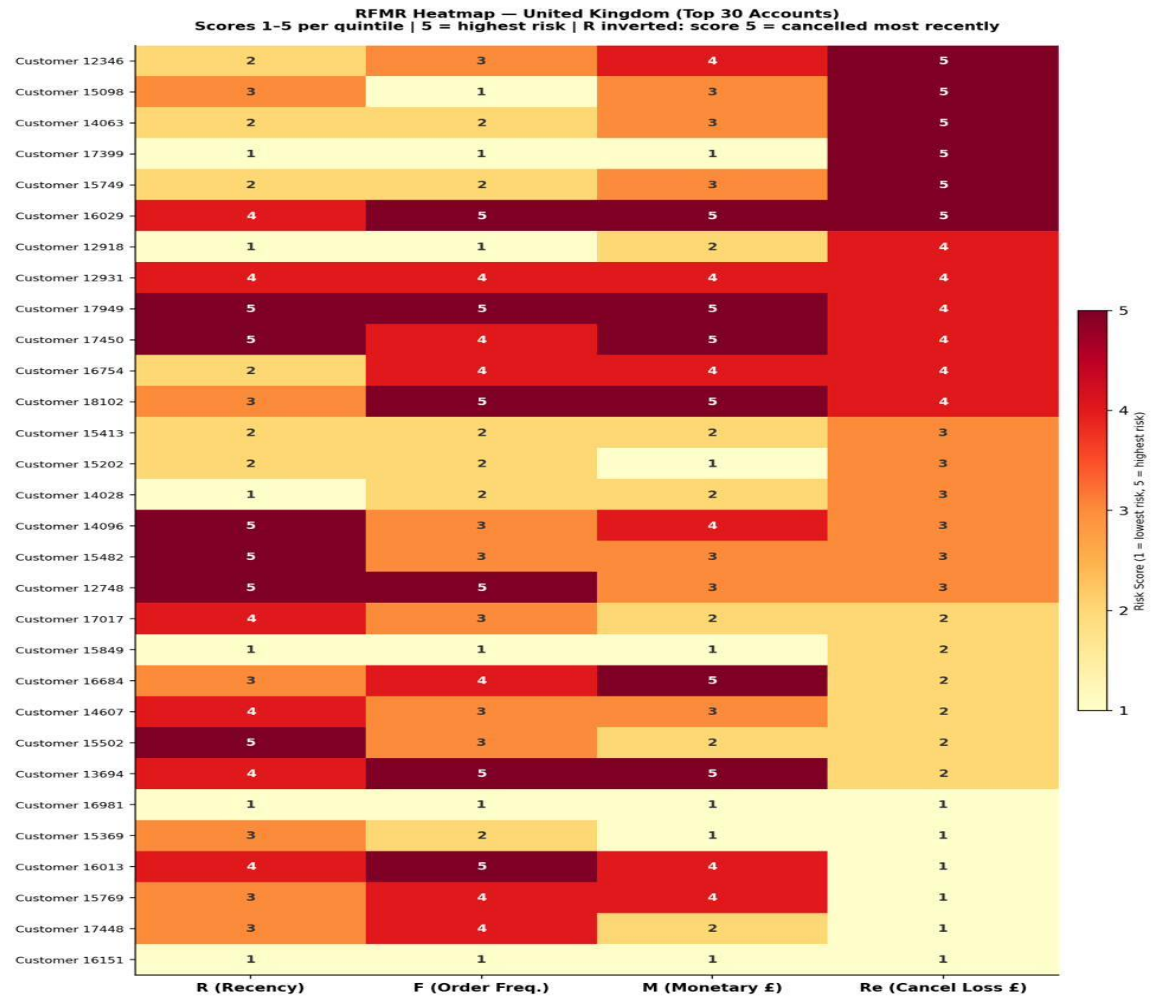

Figure 12: RFMR Heatmap — United Kingdom (Top 30 Accounts)

For example, among U.K. accounts with the largest cancellation losses (Re = 5), recency ratings for these accounts are typically 1–2. These customers have been canceling for some time now and have gone silent. For these same accounts, frequency ratings are generally 1–3, and monetary ratings are usually 3–4. Consistently, high-risk U.K. accounts are typically mid-size to large customers who have had one large cancellation event and subsequently stopped buying.

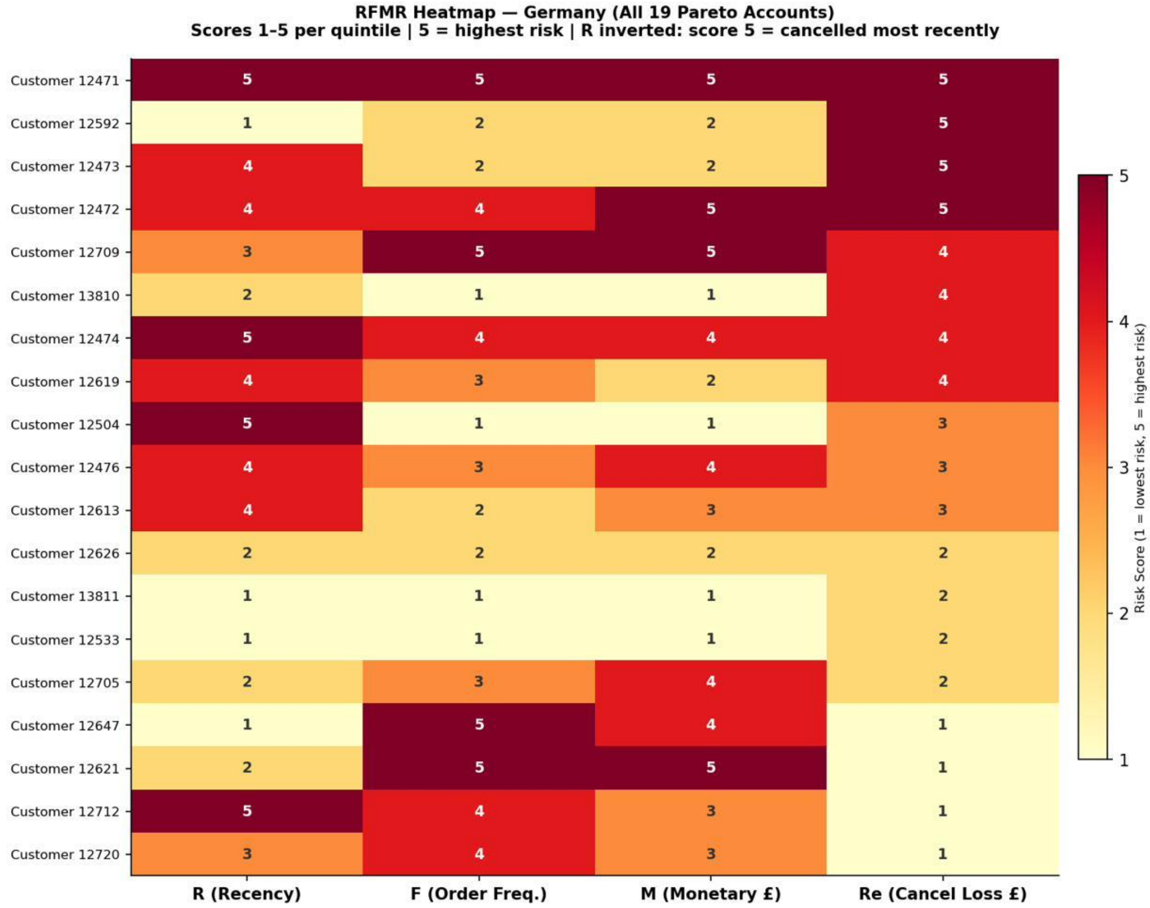

Figure 13: RFMR Heatmap — Germany (All 19 Pareto Accounts)

In contrast to the U.K., Germany shows two distinct profiles within the Re score 5 group:

- **Profile 1 One-Time Disengaged Buyer** (R: 4-5, F: 2, M: 2): These are recent, infrequent, low-spending customers who have canceled and made no further effort to buy anything else from us. We believe this is likely due to a product-market fit issue.
- **Profile 2 High-Value Chronic Canceller** (R: 4-5, F: 4-5, M: 4-5): These are frequent, high-spending customers with ongoing cancellation patterns. These customers could be experiencing operational friction or engaging in speculative bulk ordering.

Business Insight: For many UK customers who have experienced high loss levels, these losses generally stem from "past" events; as such, the key difference among those customers is the dollar size of their orders, not their behavior. On the other hand, Germany continues to experience problems with its high-loss customers, with two distinct groups that need to be treated differently. The recommended next step is to perform K-Means clustering on the complete set of UK and German Pareto accounts to verify these segments formally.

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Question 3: Is there a correlation between purchase frequency and return rate?

To investigate whether repeat business influences the likelihood and magnitude of cancellation, a Spearman's Rank Correlation coefficient was calculated to assess the relationship between Total Purchase Frequency (F) and Return Rate for Pareto Accounts across two countries, the U.K. and Germany. In this case, we used the

Return Rate Ratio rather than the absolute amount of cancellation loss, since we want to analyze the proportional nature of customer cancellation behavior rather than the absolute level of cancellation loss. To preserve any behavioral differences that emerged from Question 2, separate analyses were performed for each country. In both the U.K. and Germany, statistically significant negative correlations emerged: Spearman's Rho r = –0.804 (p = 0.0000) in the U.K. and r = –0.729 (p = 0.0004) in Germany.

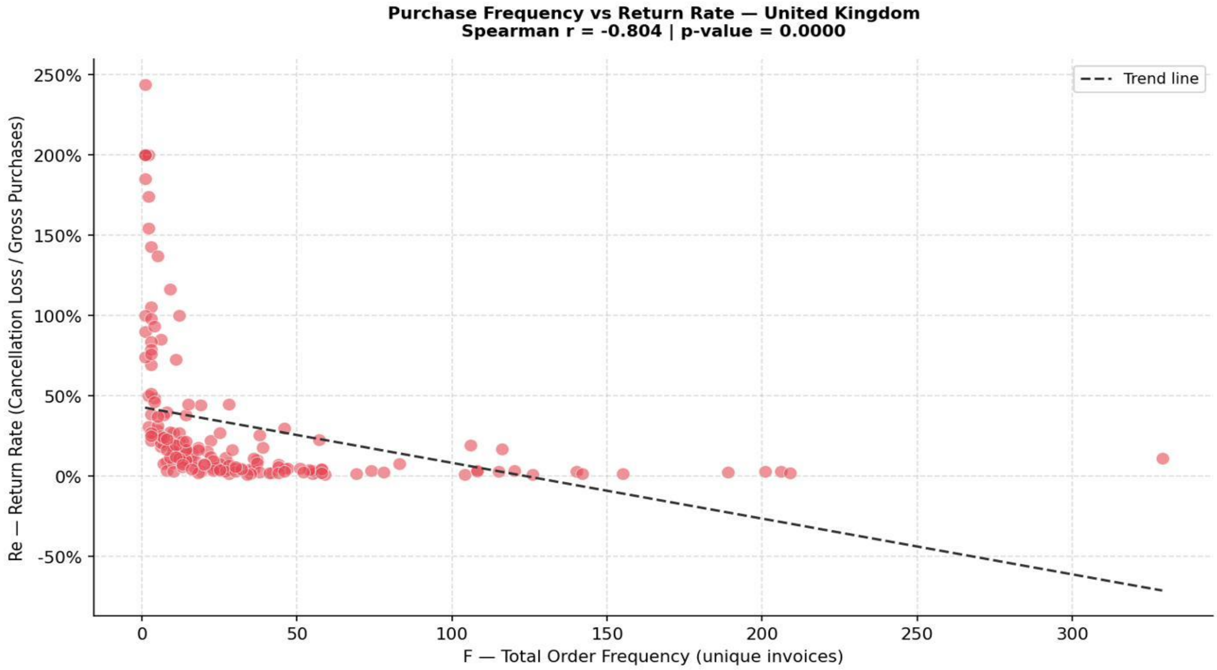

Figure 14: Purchase Frequency vs Return Rate — United Kingdom

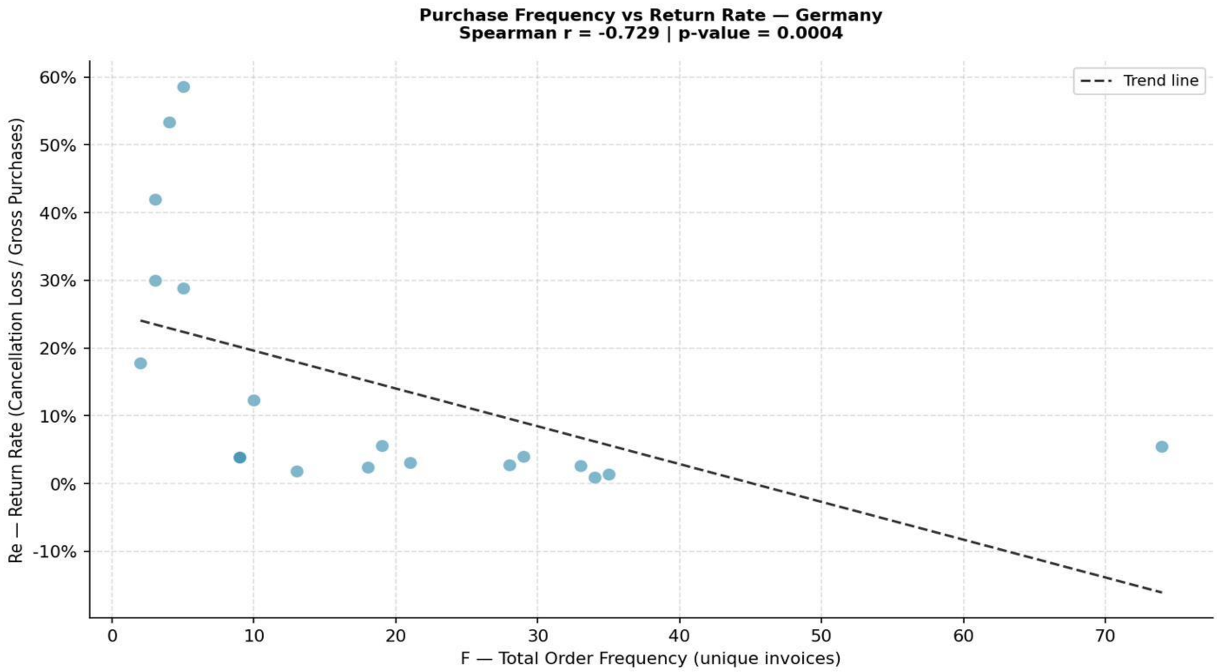

Figure 15: Purchase Frequency vs Return Rate — Germany

Scatter Plots reveal an L-Shaped Distribution in both Markets. Customers who make few purchases exhibit an unstable, highly variable range of Return Rates. However, as customers increase their purchasing frequency, their Return Rates converge towards Zero and become stable. Thus, the L-Shape is an empirical finding: customers who buy repeatedly will simultaneously reduce and stabilize their cancellation-loss behavior.

There exists a small number of customers whose Return Rates have been measured at greater than 100%, with one account exhibiting a Return Rate close to 250%. These high percentages occur when the cancellation loss is reported to be greater than the customer's gross purchases. For example, when there are discrepancies in reporting times between when orders were placed and when cancellations were noted/registered, or when StockCode M entries artificially inflate the cancellation side without corresponding purchase records. Regardless of the specific cause, these edge cases do not alter the general finding presented above.

Business Insight: High-Frequency Buyers cancel proportionately less, while Low-Frequency Buyers are in the Danger Zone since they tend to cancel at higher rates and are generally less predictable. The observed L-Shaped Distribution supports the conclusion that placing customers into Higher vs. Lower frequency categories will result in lower cancellation risk. Therefore, tighter Order Controls should be directed towards low-frequency and/or First-Time B2B Buyers.

Product Analysis

Question 1: Which products have the highest return rate as a share of units sold?

Stock Code M entries marked “Manual” in the provided data have been excluded from this portion of the product analysis since they can’t be linked to particular items. It appears these Manual Stock Codes refer to financial changes made by the vendor themselves, for example, correcting invoices or settling credit outside of normal ordering procedures. If included within this section, it would artificially inflate product cancellation amounts and obscure the ability to discern what products are being canceled. Following the removal of these Manual Stock Code entries, there are 1,039,238 product-level records, 18,473 cancellations across 2,876 individual SKUs, and a total cancellation loss of £583,236.

Although Return Rate and Loss Amount provide differing insights into the business performance, both need to be evaluated. When evaluating Loss Amount only, the top item in loss is the Medium Ceramic Top Storage Jar (SKU #23166), with a total loss of £77,480, or 13.28 percent of all product cancellation loss. However, when evaluating Return Rates, the story changes dramatically. For example, the Vintage Blue Vacuum Flask has returned at 100% while the Black Cherry Lights has returned at 50.5%. Therefore, while the ceramic jar accounts for the highest dollar amount lost to returns, it may actually be performing better than some other products that return at 100%.

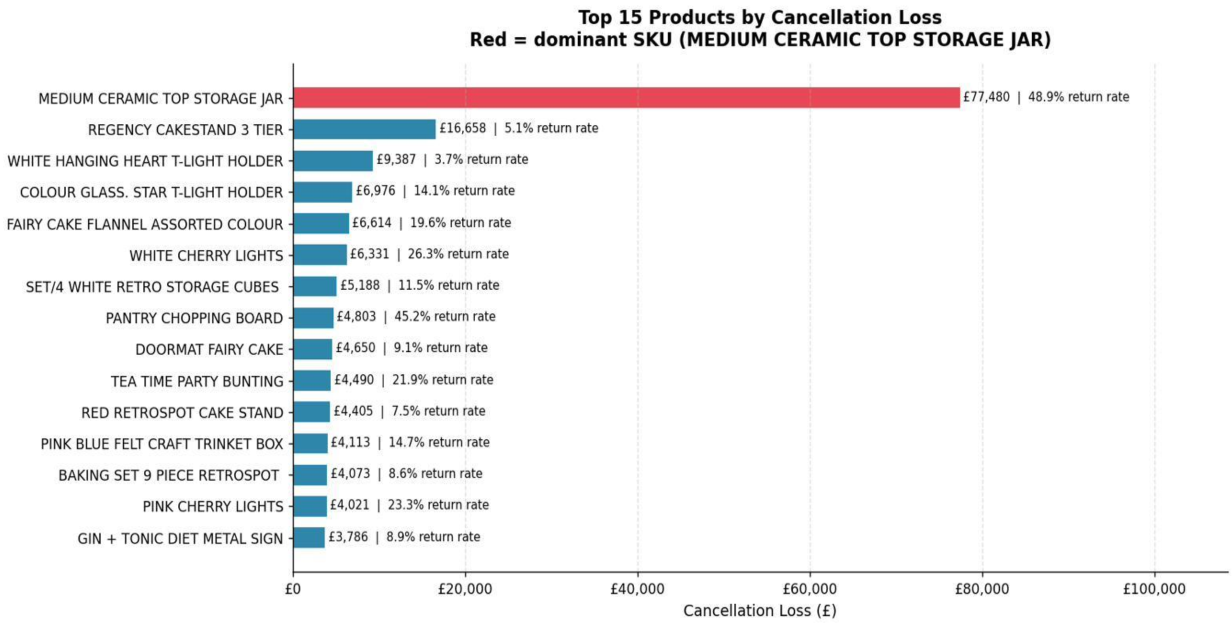

Figure 16: Top 15 Products by Cancellation Loss

When looking more closely at the ceramic jar, we see that nearly all of the issues can be explained by a single event. Customer number 12346 canceled 74,215 of their ceramic jars in January of 2011 in one transaction worth £77,183. Before January of 2011, the SKU did not exist. The SKU was introduced and immediately canceled by customer number 12346. Since then, each month's return rate has hovered around the low-five-percentile range.

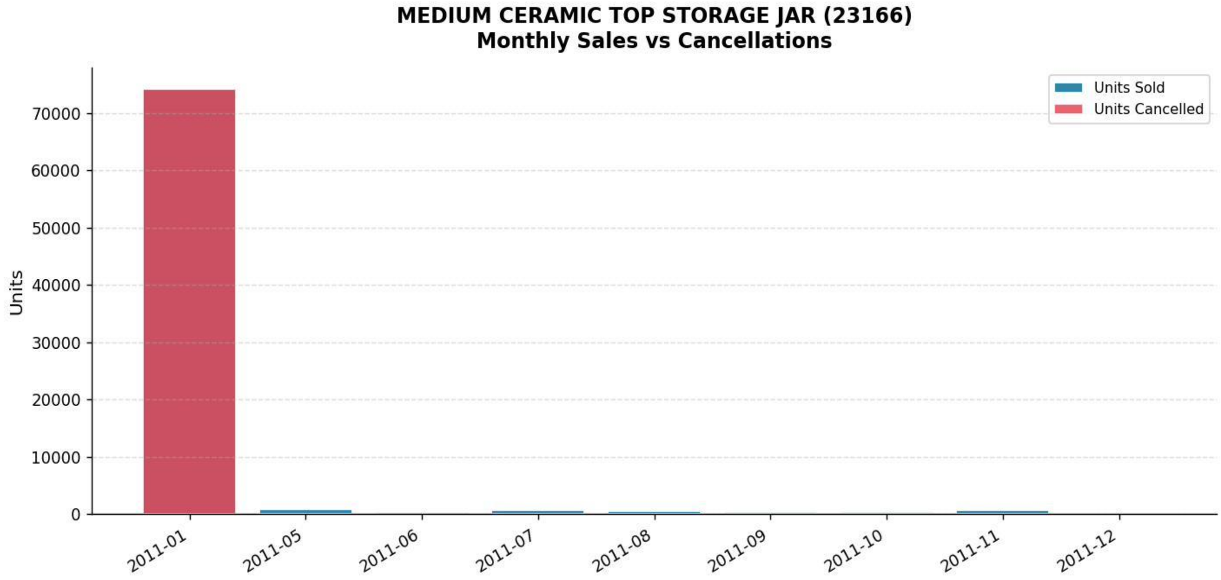

Figure 17: Medium Ceramic Top Storage Jar (23166) — Monthly Sales vs Cancellations

As stated above, the same applies to the September 2010 spike of 95,156 canceled units. Of those 87,167 (or 91.6%), were from a single invoice placed by customer number 14277 located in France. That single invoice was for a large wholesale purchase placed across approximately 20 mixed SKUs in a single transaction and was subsequently canceled for those SKUs. There is no indication of a pattern of product issues.

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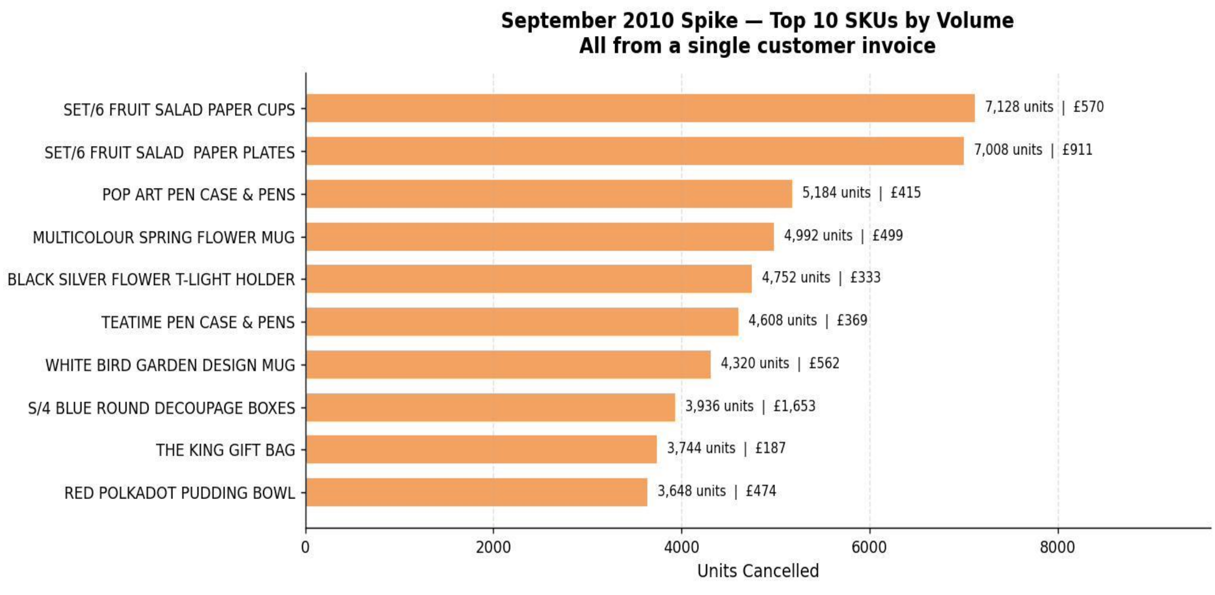

Figure 18: September 2010 Spike — Top 10 SKUs by Volume

Business Insight: Before making decisions about any SKU that is either a high loser or a high returner, verify whether the trend will continue or is simply a one-off. At first glance, the Ceramic Jar seems to be a very poor performer at 48.9% and £77,480, but the data indicate it is simply a single customer and an invoice from a single month. Similarly, products that return at 100%, such as the Vintage Blue Vacuum Flask and the Set 6 Mini Sushi Set Fridge Magnets, should also be reviewed from a financial perspective. A product that generates zero net positive sales over its entire existence is a cost center rather than a source of revenue for the organization.

Question 2: Do high-return products share characteristics such as price range or category?

The answer will depend upon the metric used. In terms of absolute cancellation loss, the £1-3 pricing range accounts for 47.9% of all product cancellation loss (£279,211), as many products fall into this category; therefore, many cancellations occur in this range. However, return rates tell a different story. On average, return rates for items priced under £1 are 3.1%, rising to 15.7% for items priced over £50. Therefore, regardless of the number of individual items ordered, and whether these orders have been placed at low prices (and thus represent more cancellations), the higher an item costs, the greater its per-unit value in terms of potential losses due to return rates.

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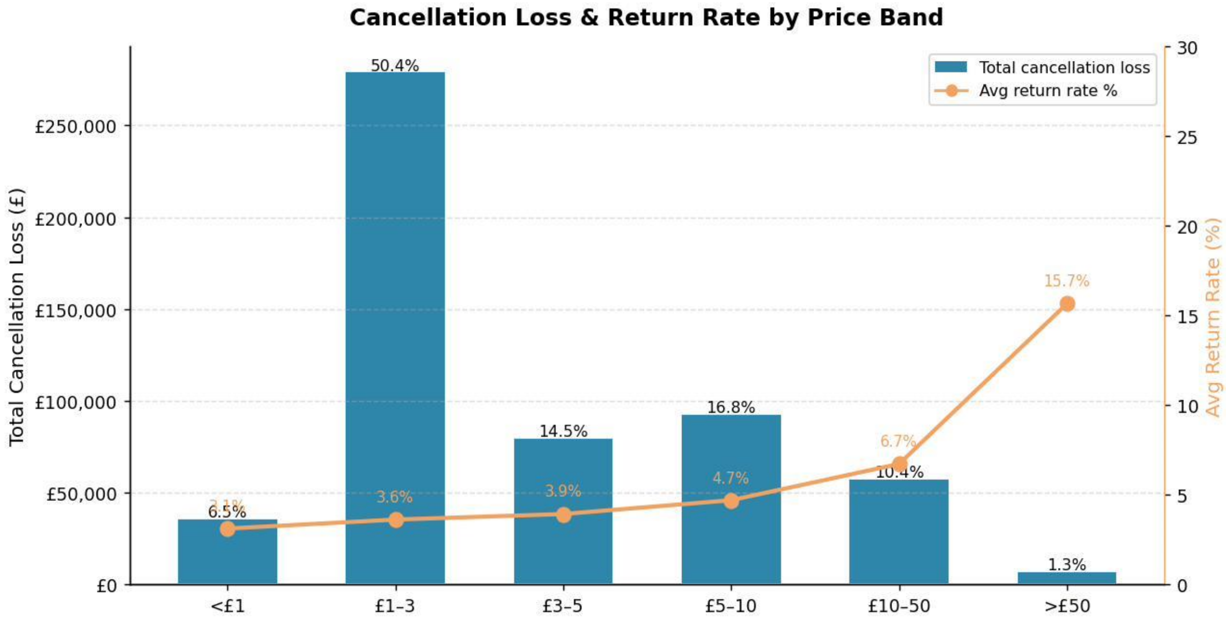

Figure 19: Cancellation Loss and Return Rate by Price Band

Business Insight: Products priced cheaply lose money through their volume; products priced more expensively lose money through their return rates. As such, volume-driven losses in the £1-3 price range should be addressed by implementing changes to product descriptions, packaging, and related elements. Quality expectations or issues associated with order fulfillment may need to be investigated further if return rates for more expensive products continue to remain elevated.

Question 3: Does a Pareto pattern hold? Do a small number of products account for most returns?

Yes. Of the 3058 SKUs that have been canceled, 438 (14.3%) accounted for 80% of cancellations by dollar value and can explain the rest. The data follow the same trend as the geography comparison (where a single country accounts for 85% of losses); however, the distribution of canceled products is significantly broader than that of geographies. Therefore, individual SKU management will likely be less effective than managing products by category or price.

There is still some distortion at the upper end of this distribution due to the previously mentioned account-related events in Q#1, like the Ceramic Jar and the September 2010 spike. When these events are excluded from consideration, the remaining concentration of cancellations is legitimate; however, they remain sufficiently dispersed to make individual SKU management less optimal than grouping by category or price bands.

In terms of the four Pareto countries: Eire, Norway, France, and Spain, there was an existing relationship established through the customer analysis as being "account-driven" issues. This provided a framework to test whether a product signal existed beneath the account behavior observed in each country.

- **Norway:** Once StockCode M Entries are eliminated, all cancellation activity is essentially gone. There does not appear to be a product issue.
- **France:** All cancellation activity is attributed solely to a one-time, large-volume transaction by customer #14277 in September 2010. There appears to be no underlying SKU issue.
- **EIRE:** Legitimate product signal. The Regency homewares/tea service SKU set, Cakestand 3 Tier, Roses Teacup and Saucer, Large Cake Stand Hanging Strawberry, Teapot Roses, Sugar Bowl Green — continues to cancel over time.
- **Spain:** Customer #12454 has been canceling the same home decor SKUs: Love Garland, French WC Sign, Wooden Happy Birthday Garland, over multiple invoices in like quantities. While there may be other customers canceling these SKUs, they represent the most extreme example of the product-cancellation pattern observed in this dataset.

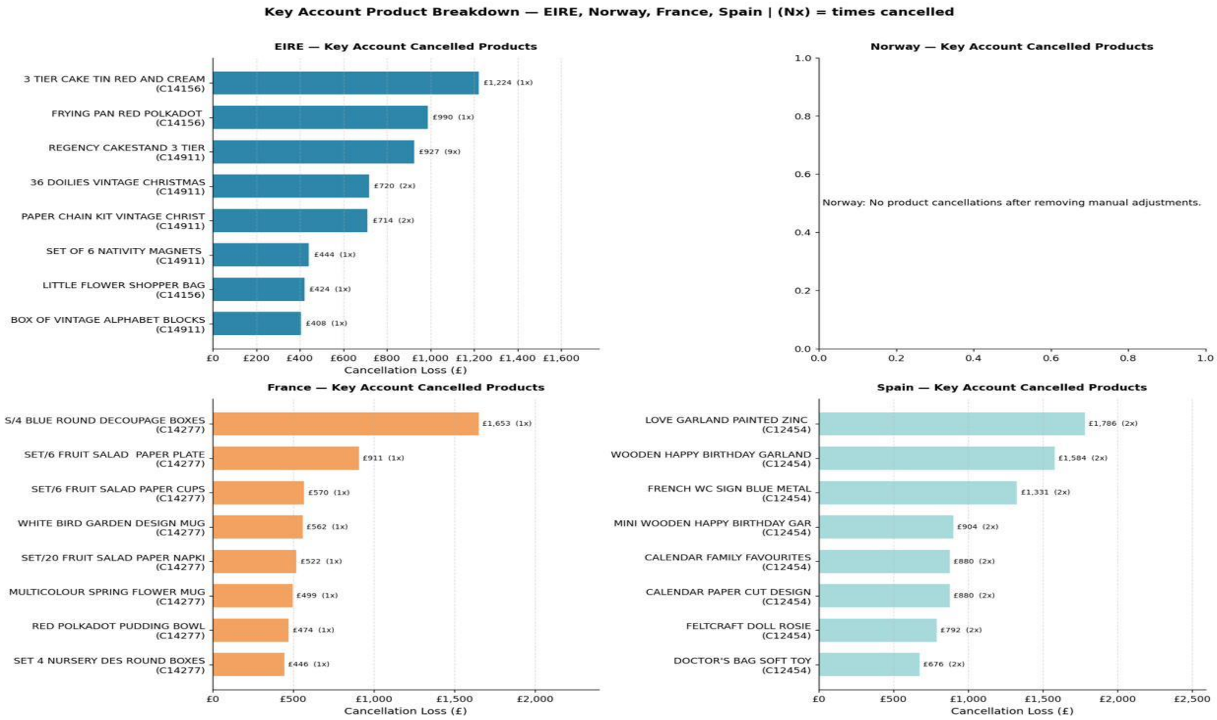

Figure 20: Key Account Product Breakdown — EIRE, Norway, France, Spain

Business Insight: The Pareto data cannot be viewed in isolation. Top-of-the-distribution (account) events, such as product issues, will need to be resolved at the account level rather than with a product solution. Genuine product signals exist within Eire's Regency line and Spanish home decor categories. Account-level segmentation will be needed to accompany product solutions for the UK & Germany. There is no singular product strategy that can support all six countries.

Business Interpretation and Recommendations

Geographic and Temporal

- Focus the intervention on the top six countries, which represent 95% of total cancellation losses. The UK accounts for 85%, so it is both the priority and the most complex market to address.

- Preposition logistics and customer service resources before Q4 and the post-holiday period in Q1; the seasonal volume surge is predictable.

Customer

- **EIRE, Norway, France, and Spain:** direct account management. Review the order history for each Pareto account and implement order-confirmation protocols or deposit requirements for large B2B orders. Fewer than 10 accounts across all four markets account for the majority of the combined accounts.
- **Germany:** Segment before intervening. With 19 accounts needed to reach 80% and no dominant player, a one-size-fits-all policy is inefficient. Identify shared behavioral patterns first.
- **UK top accounts:** individual review. These are medium-to-large buyers who canceled once and went quiet. Minimum order thresholds and staged payment terms for large one-off b2b orders reduce future exposure without affecting loyal frequent buyers.
- **Germany Profile 1 (one-time disengaged):** investigate product-market fit. These customers tried, canceled, and left. Review whether product descriptions or category offerings are creating unmet expectations for first-time buyers in the German market.
- **Germany Profile 2 (high-value chronic cancellers):** Direct account engagement. These are worth retaining. Outreach to understand recurring cancellation drivers could convert chronic cancellers into stable, high-value accounts.
- The L-shaped correlation finding supports three cross-market actions: invest in loyalty incentives to retain frequent buyers; Apply tighter controls to infrequent and first-time b2b buyers where return rates are highest; use purchase frequency as an early warning indicator when flagging large one-off orders for pre-shipment review.

Product

- Investigate the Regency product line in EIRE directly. Several months of consistent cancellations across a coherent product family cannot be explained by account behavior alone. Review quality, product descriptions, and whether there is a fulfillment issue specific to that market.
- Investigate Spain's home décor cancellation pattern before scaling that product line. Customer 12454 canceling the same SKUs across separate invoices suggests either unmet expectations or a fulfillment issue specific to those items.
- Apply order confirmation or deposit requirements for large one-off B2B orders of high-priced SKUs. Items above £50 have a return rate of 15.7%, five times the rate for cheap items.
- Conduct a financial viability review of products with a 100% return rate: Vintage Blue Vacuum Flask, Set 6 Mini Sushi Set Fridge Magnets. A product with a 100% return rate becomes a cost center.

Limitations and Future Work

Limitations

Geographic Analysis

All data has been recorded at the country level only. Therefore, there is no city-, regional-, or postal-code data available to measure whether individual cities within countries contribute to higher-than-expected loss rates. Additionally, geographic proximity (i.e., distance) to the retailer’s distribution center may have indirectly contributed to order cancellations, but this cannot be confirmed.

Temporal Analysis

The time frame of the data set does not represent a complete two-year window. Although the data set begins on December 1st, 2009, the end date is December 9th, 2011, which is partway through the month. Only 2010 and 2011 represented full calendar years. Due to the limited number of quarterly patterns observed during quarter four, seasonality cannot be concluded with statistical certainty.

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Customer Analysis

The RFMR scores are computed separately for each country group. Therefore, comparisons among countries are not possible. A value of five for a German customer does not equate to a value of five for a UK customer in absolute terms. The two behavioral profiles developed in Germany are descriptive statements and not formally defined segments. Cluster analysis utilizing K-Means will be needed to validate them.

Product Analysis

The use of Stock Code M in some entries introduces ambiguity into the data set. As such, the percentage of cancellations that reflect actual product returns and the number of different product types involved cannot be determined from the existing data. Because of this limitation, Norway's product-related narrative may never be able to be resolved using the existing data. There is no additional categorization of product categories beyond the stock code classification, nor was there sufficient information regarding suppliers. These limitations prevent a conclusion about whether product categories or suppliers drive cancellations.

Future Work

- Validation of the behavioral segments identified within the customer analysis via K-means clustering based on the four RFMR dimensions (as inputs) across the entire 190 U.K. & 19 German Pareto accounts. This will provide formal validation and also provide scalable opportunities for intervention.
- Review of each Pareto customer account one by one in Ireland, Norway, France, and Spain. Due to the low volume of accounts and the level of individual exposure, this is the most effective way to understand the underlying reasons for cancellations.
- Spatial sub-national data, including city/post code, may help to investigate if logistical distances and/or regional issues are contributing factors to cancellations, especially within the U.K.
- Longer time series data. A dataset spanning four to five years would provide an opportunity to confirm seasonality through statistical methods rather than infer it from the two Q4 observations made during the study.
- Category and Supplier Enrichment. Adding category classification beyond product name and supplier identifier will enable further investigation into whether categories or suppliers drive the risk of cancellation.
- Return Reason Codes. Capturing why customers cancel closes the inferential gap across all three sections, as previously mentioned. In other words, although patterns have been identified, we cannot explain them.
- Viability Review of Products with 100% Return Rates. The business needs to determine if canceling the above-mentioned items (Vintage Blue Vacuum Flask, Set 6 Mini Sushi Set Fridge Magnets) provides greater value to the business than continuing to stock them and incur additional costs related to reverse logistics.

References

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AI declaration

Claude (Anthropic, 2025) was an artificial intelligence tool employed to support text editing, add humanism to writing style, and assist in developing analytical storylines. The authors make all decisions regarding analysis, interpretation, and conclusion.